## About the project

This project was developed by Emilio Barragán, a Computer Science student at Tecnológico de Monterrey, as part of a scholarship research program conducted from August to November 2024. During the project, while in my third semester, I collaborated with six other students under faculty supervision.

This project uses Natural Language Processing (NLP) and supervised machine learning to classify evaluations written by students about their professors.
The main goal was to identify between 0 and 2 labels per negative or positive aspect—for example, teaching method, attitude, and effectiveness. With these labels, create a machine learning model using traditional algorithms to predict these labels.

## About the dataset

As mentioned, the dataset contains evaluations written in Spanish by students from Tecnológico de Monterrey about the quality and positive or negative aspects of their teachers. The dataset had slightly more than 6000 reviews, and together with 6 other students, I manually had to identify the positive and/or negative aspects of the teacher just by reading the review. These positive or negative aspects were the labels mentioned, and there were only 9 possible labels (including one for neutral cases):
Teacher-student relationship, Teaching attitude, Teaching method, Not mentioned, Classroom atmosphere, Teacher image, Teaching effectiveness, Teacher quality, Teaching content, and Teaching ability.

Since we had to identify positive and/or negative aspects, if there were no positive or negative aspects mentioned in the review, the label was going to be 'not mentioned'. Additionally, there was a maximum of 2 labels; this meant that there could be 'Teacher image' and 'Teaching ability' as a positive or negative aspect.

To train and create the model, I had to do text preprocessing, which included removing stopwords, cleaning the text (removing punctuation marks, capital letters, and extra spaces), and vectorizing the data so that the algorithm could process and learn from it.

## Methodology

The project began with an initial exploration of the dataset to understand the distribution of labels, the length of the reviews, and the frequency of certain words or expressions related to teaching performance.

Data cleaning: punctuation removal, lowercasing, and whitespace normalization, removing fake or mispelled labels.

Stopword removal using NLTK’s Spanish stopword list. (Which I imported a txt file)

Vectorization of text to transform the reviews into numerical features suitable for machine learning algorithms. Encoding of multiple labels using MultiOutputClassifier from scikit-learn.

Since the task was multi-label classification, the evaluation focused on metrics such as F1-score (macro and micro), Exact Match Ratio, and Hamming Loss.
These metrics allowed for assessing both the accuracy of label prediction and the model’s ability to handle multiple classes simultaneously.

## Coding and packages

The coding language used in this project was mainly Python in a Jupyter Notebook environment. For the machine learning part, I used scikit-learn to train and split the model and to evaluate the metrics, mainly the F1 score, as well as NumPy and Pandas to visualize and use the dataset (not graphics).

## Results

| | precision | recall | f1-score | support |
|---|---|---|---|---|
| 0 | 0.00 | 0.00 | 0.00 | 75 |
| 1 | 0.80 | 0.02 | 0.03 | 239 |
| 2 | 0.00 | 0.00 | 0.00 | 208 |
| 3 | 1.00 | 0.01 | 0.01 | 178 |
| 4 | 0.00 | 0.00 | 0.00 | 40 |
| 5 | 0.00 | 0.00 | 0.00 | 2 |
| 6 | 0.50 | 0.00 | 0.01 | 286 |
| 7 | 0.76 | 0.52 | 0.62 | 541 |
| 8 | 0.62 | 0.02 | 0.04 | 263 |
| 9 | 0.00 | 0.00 | 0.00 | 188 |
| micro avg | 0.76 | 0.14 | 0.24 | 2020 |
| macro avg | 0.37 | 0.06 | 0.07 | 2020 |
| weighted avg | 0.54 | 0.14 | 0.18 | 2020 |
| samples avg | 0.24 | 0.13 | 0.17 | 2020 |

Since this classification report does not directly have an exact match ratio, we can assume it's low since the micro recall is 0.14 and the samples avg is 0.17, the exact match ratio could be between 0.13 and 0.15. However, the macro score was included in this report, and it's 0.07, which is very low, but expected since the dataset has some labels that don't repeat that much, making it hard for the model to make predictions on them.

## Conclusions

Despite getting a report with unexpected results, there were some good results; for example, despite the F1-score being 0.07, there was a label that had a score of 0.62. This means that this label had more sample data to train the model than other labels, and this could be solved with a more balanced splitting of the dataset, which in this case was done randomly by scikit-learn. The size of the test data was 20 percent of the whole dataset.